MIL-C-62279A(AT)
4 October 1985
SUPERSEDING
MIL-C-62279(AT)
AMENDMENT 1
18 May 1979

MILITARY SPECIFICATION

CHASSIS, TANK; ARMORED VEHICLE, BRIDGE LAUNCHER, M60A1 AND M48A5; PROCESSING FOR STORAGE AND SHIPMENT OF

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the processing of the M60A1 and M48A5 tank chassis for shipment and storage.

1.2 Classification. Processing shall be of the following levels as specified (see 6.2).

Level A	Processing for domestic or oversea shipment and outside storage in excess of 90 days from date of processing (periodic care and preservation during storage required).
Level B	Limited processing for immediate use shipment and for domestic or oversea shipment (excluding open deck loading) and any storage not to exceed 90 days from date of processing.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: AMSTA-GSS, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2350

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS
FEDERAL

A-A-1685	- Tape, Pressure-Sensitive Adhesive Black, Weather-Resistant, Glass Filament Reinforced.
O-S-576	- Sodium Bicarbonate, Technical.
O-S-801	- Sulfuric Acid, Electrolyte for Storage Batteries.
P-D-220	- Detergent, General Purpose.
NN-P-530	- Plywood, Flat Panel.
QQ-A-250	- Aluminum Alloy 6061, Plate & Sheet.
QQ-W-461	- Wire, Steel, Carbon (Round, Bare and Coated).
RR-W-360	- Wire Fabric, Industrial.
TT-E-527	- Enamel, Alykd, Lusterless.
TT-E-529	- Enamel, Alykd, Semi-Gloss.
TT-V-121	- Varnish, Spar, Water-Resisting.
UU-T-81	- Tags, Shipping and Stock.
VV-F-800	- Fuel Oil, Diesel.
VV-L-800	- Lubricating Oil, General Purpose, Preservative (Water-Displacing, Low Temperature).
ZZ-T-416	- Tire Rebuilding and Tire and Tube Repair Materials.
MMM-A-178	- Adhesive, Paper Lable, Water-Resistant.
MMM-A-1617	- Adhesive, Rubber Base, General Purpose.
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Locked Corner.
PPP-B-636	- Box, Fiberboard.
PPP-C-843	- Cushioning Materials, Cellulosic.
PPP-C-1752	- Cushioning Material, Packaging, Unicellular, Polyethylene Foam Flexible.
PPP-C-1797	- Cushioning Material, Resilient, Low Density, Unicellular Polypropylene Foam.
PPP-T-60	- Tape; Pressure Sensitive Adhesive, Waterproof - for Packaging and Sealing.

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MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-B-117	- Bags and Sleeves, Interior Packaging.
MIL-C-450	- Coating Compound, Bituminous, Solvent Type, Black.
MIL-C-5501	- Cap and Plug, Protective, Dust and Moisture Seal.
MIL-A-8421	- Air Transportability Requirements, General Specifications for.
MIL-P-12841	- On Vehicle Equipment (OVE) for Military Vehicles Packaging of.
MIL-P-13983	- Paint, Temporary, Lusterless, Gasoline Removable.
MIL-E-16053	- Electrodes, Welding, Bare, Aluminum Alloys.
MIL-C-16173	- Corrosion Preventive Compound, Solvent. Cutback, Cold-Application.
MIL-D-16791	- Detergents, Nonionic.
MIL-L-21260	- Lubricating Oil, Internal-Combustion. Engine, Preservative.
MIL-T-22085	- Tape, Adhesive, Preservation and Sealing.
MIL-F-22191	- Films, Transparent, Flexible, Heat-Sealable, for Packaging Applications.
MIL-W-45205	- Welding, Gas Metal-Arc and Gas Tungsten-Arc, Aluminum Alloys Readily Weldable for Structures, Excluding Armor.
MIL-L-46002	- Lubricating Oil, Contact and Volatile Corrosion Inhibited.
MIL-P-46167	- Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-D-81298	- Dye, Liquid Red, for Detection of Leaks in Aircraft Fuel Systems.

STANDARDS
MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MS16562	- Pin, Spring - Tubular, Slotted.
MS27040	- Nut, Plain, Square-Steel, Cadmium Plated.
MS27183	- Washer, Flat-Round, Steel, Cadmium Plated, General Purpose.
MS34338	- Washer, Lock-Spring, Helical, Regular (Medium) Series.
MS35495	- Screw, Wood, Round Head, Slotted, Steel and Brass.
MS35751	- Bolf, Square Neck, Flat Head (Elevator) Steel, Cadmium or Zinc Plated.
MS35842	- Clamp, Hose, Low Pressure, Type F.
MS90726	- Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel, Grade 5, Cadmium Plated, UNF-2A.

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2.1.2 Other Government documents, drawings, and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS
ARMY

10870861	- Packing, Performed Rubber.
K10893902	- Installation Vehicle Protective Closure.
11655238	- Finger.
11655239	- Flange, Closing Plate.

PURCHASE DESCRIPTION

M60PD-T-62226(AT)	- Vehicle Closure.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASSOCIATION OF AMERICAN RAILROADS PUBLICATION

Section No. 1	- General Rules Governing Load of Commodities on Open Top Cars.
Section No. 6	- Rules Governing the Loading of Department of Defense Materiel on Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 59 East Van buren, Chicago, Illinois 60605).

AMERICAN WELDING SOCIETY

AWS A5.10	- Aluminum and Aluminum Alloy Base Welding Rods and Electrodes, Specification for.

(Application for copies should be addressed to the American Welding Society, Inc., 550 N.W. Le Jeune road, P.O. Box 351040, Miami, Florida 33135).

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2), one of the first ten production processed chassis shall be subjected to the inspection specified in 4.3. Approval of this chassis shall not relieve the contractor of his obligation to process all chassis in accordance with this specification. Unless otherwise specified by the Government, any changes to materials or design after approval shall require an additional chassis to be inspected as specified in 4.3. The Government representative shall select the vehicle to be inspected.

3.2 Materials. Materials shall be as specified herein and in referenced specifications and standards. Materials shall be free of defects which adversely affect performance or serviceability (see 4.5.1).

3.3 Processing levels. Level A processing shall be as specified in 3.4. Level B processing shall be as specified in 3.5 and shall constitute exceptions or additions to the processing in Level A (see 4.5.2).

3.4 Level A.

3.4.1 Disassembly. Parts vulverable to damage and pilferage, and projecting parts whose removal will accomplish the desired reduction in cube, shall be removed from the chassis. Removed parts shall be preserved, packaged, and packed in accordance with the requirements for the level to which the chassis is being processed, and the applicable specifications. The packed parts shall be placed in a protected location in the chassis an secured in a manner to prevent movement and damage during shipment and storage. Removed bolts and nuts, screws, pins, and washers shall be placed in one of the mating parts and secured to prevent their loss.

3.4.2 Matchmarking. Parts removed from the chassis shall be matchmarked when necessary to facilitate reassembly. Matchmarking information shall be on cloth shipping tags conforming to type A of specification UU-T-81, or metal tags marked with soluble paint and attached to mating parts. The marked cloth shipping tags shall be waterproofed with varnish conforming to TT-V-121 or adhesive conforming to MMM-A-178.

3.4.3 Record forms. Two copies of DD Form 1397 shall be provided. Information on forms shall include preservation accomplished and depreservation instructions. The Equipment Log Book Binder and one copy of DD Form 1397 shall be placed in a bag conforming to type I, style 2 of MIL-B-117; bag shall be closed by heat sealing and securely attached inside the vehicle. The other copy of DD Form 1397 shall be waterproofed with adhesive conforming to MMM-A-178 and securely attached in a conspicuous location on the exterior of the vehicle (see 4.5.2).

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3.4.4 Cleaning and drying.

3.4.4.1 Vehicle Interior. Chassis interior surfaces shall be cleaned in accordance with process C-1 of MIL-P-116, except that no liquids under pressure shall be used in any compartment other than engine compartment. Cleaned surfaces shall be dried by any applicable procedure specified in MIL-P-116.

3.4.4.2 Battery supports and retainers. Battery supports and retainers shall be cleaned with a solution composed of 1/2 pound of sodium bicabonate, conforming to O-S-576, per gallon of water, and shall be flushed with clean water, then thoroughly dried. Dried surfaces shall then be preserved in accordance with 3.4.5.2.

3.4.4.3 Vehicle Exterior The vehicle exterior shall be cleaned using a solution of detergent conforming to P-D-220, or type I of MIL-D-16791, in water or steam. Cleaning shall remove all foreign matter. Cleaned surfaces shall be rinsed with clean water or steam and thoroughly dried. Care shall be taken to avoid entry of water or steam into the engine compartment, personnel heater exhaust tube, air cleaner ports, or other chassis openings.

3.4.5 Preservation.

3.4.5.1 Relubrication. When the chassis has been operated in excess of 50 miles since previous lubrication, or after a detergent or steam cleaning, chassis shall be lubricated using materials in accordance with drawings, specifications, or lubrication order pertinent to the chassis. All exposed oil can points such as levers, locking bars, strikers, hinges, hinge pins, lockings pins, pintle pins, locking levers, wing nuts, latches, door locks, control rod clevis and pins, brake lever linkage, engine throttle control linkage, and transmission support guide rails, shall be lubricated with oil conforming to VV-L-800. Excess oil shall be removed after lubrication.

3.4.5.2 Battery supports and retainers. Battery supports and retainers shall be preserved with compound conforming to MIL-C-450.

3.4.5.3 Transmission and final drives. The transmission shall contain lubricating oil only conforming to type I, grade 10 of MIL-L-21260 filled to the operating level. Final drives shall contain lubricating oil conforming to type I, grade 10 or 30, as applicable, of MIL-L-21260 filled to operating level. DD Form 1397 shall be annotated with type and grade of lubricant used.

3.4.5.4 Engine crankcase. The engine crankcase shall be filled to operating level with lubricating oil conforming to type I of MIL-L-21260 of the seasonable grade specified in the applicable drawing, specification, or lubrication order. DD Form 1397 shall be annotated with the type and grade of lubricant used. For vehicles being prepared for shipment to, and storage in, areas where temperatures are expected to be at or below -20 degrees Fahrenheit (°F) a red warning tag containing the following instructions shall be attached in a conspicuous location in the driver's compartment: "DRAIN ENGINE CRANKCASE AND REFILL WITH MIL-L-46167 OIL BEFORE OPERATING ENGINE" (see 6.2).

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3.4.5.5 Engine preservation. The engine shall be preserved in accordance with 3.4.5.5.1 through 3.4.5.5.5.

3.4.5.5.1 Fuel system and combustion chamber. The fuel system and combustion chamber shall be processed in the following steps.

Step 1. Prior to processing the engine shall be cooled to assure that the cylinder head temperature, measured at the injector nozzle flange surfaces of all cylinders, is not more than 100°F. Cooling shall be accomplished by induced air currents or by waiting the period of time required to arrive at the specified temperature. When the ambient temperature exceeds 100°F, the engine shall be cooled to the ambient temperature. After the engine has been cooled, the fuel supply from the fuel tanks shall be shut off.

Step 2. A portable container with two compartments shall be positioned to provide gravity feed to the engine. One compartment shall be filled with lubricating oil conforming to grade I of MIL-P-46002 colored with an oil soluble red dye conforming to MIL-D-81298, in a concentration sufficient to impart a marked coloring to the oil. The second compartment shall be filled with diesel fuel conforming to VV-F-800. Uncouple the quick-disconnect on the inlet fuel line to the primary fuel filter. Remove filter cans and elements from both the primary and secondary filters. Drain diesel fuel from filter cans and fill them with lubricating oil conforming to MIL-P-46002, grade 1, and reinstall them without filter elements. Uncouple the engine fuel return line quick-disconnect and fasten a transparent plastic recovery line to the engine line. Provide a recovery container for fuel being discharged from the transparent line.

Step 3. Deliver MIL-L-46002 oil from the portable tank while operating the engine at idle speed, 700 revolutions per minute, for not more than 60 seconds to consume fuel in the injection system and to pump MIL-P-46002 oil into the engine fuel system. Following the 60 seconds of engine operation, the hand purge pump shall be operated without depressing the solenoid switch until lubricating oil is flowing through the transparent line into the recovery container (The change should occur after approximately two gallons of fuel flow-out).

Step 4. Remove the electrical leads from the intake manifold heater igniter plugs and operate the hand purge pump 20 strokes, with the solenoid switch depressed, to pump MIL-P-46002 lubricating oil into the intake manifold heaters. Unclamp the hose from each turbocharger inlet. Secure an air restrictor cover, fabricated in accordance with figure 1 to each turbocharger inlet. While the portable tank furnishes MIL-P-46002 oil, depress the accelerator to full control and crank the engine for three 30 second periods as follows: Cranking shall be for a period of 30 seconds; the starter shall be allowed to cool for not less than three minutes; the engine shall be cranked for an additional 30 seconds, the starter shall be allowed to cool for not less than three minutes; the engine then shall be cranked for a final period of 30 seconds. 1/ 2/

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1/ Each cranking period shall be limited between 25 and 35 seconds. Special precautions shall be taken to assure that time limits specified shall not be exceeded, as the engine, the starter, or starter solenoid may be damaged.

2/ The engine may fire for five seconds while being cranked with air restrictors installed.

Step 5. Reconnect the electrical leads to the intake manifold heater igniter plugs. Remove the air restrictor covers, but do not reconnect the turbocharger hoses (see 3.4.5.5.3).

3.4.5.5.2 Engine purging. Engine purging shall be accomplished as follows:

Step 1. The portable container shall be adjusted to provide diesel fuel for purging the return fuel line, fuel filters, engine fuel pump and injector pump. Remove the filter cans, drain MIL-L-46002 lubricating oil and wipe them clean with lint-free wiping material. Replace the filter elements in the cans, fill with diesel fuel, and reinstall the filter cans.

Step 2. Hold the engine fuel cut-off switch in the OFF position while operating the hand purge pump, without depressing solenoid switch, until diesel fuel is flowing through the transparent line into a recovery container. (This occurs after approximately two gallons flow-out of MIL-L-46002 lubricating oil.

Step 3. Disconnect recovery line and reconnect the engine fuel return line to the fuel tanks. Turn on the vehicle fuel supply system and reinstall any parts disassembled during engine processing.

3.4.5.5.3 Preservation of turbochargers. Two ounces of preservative oil conforming to MIL-P-46002, grade 1, shall be atomize-sprayed in each turbocharger inlet. The removed clamps shall be reinstalled on each turbocharger air inlet horn. Turbocharger inlets and air cleaner hoses shall be sealed with type II of MIl-T-22085 tape, or with plastic plugs conforming to MIL-C-5501.

3.4.5.5.4 Preservation through exhaust system. After preservation in accordance with 3.4.5.5.1 through 3.4.5.5.3, two ounces of preservative oil conforming to grade 1 of MIL-P-46002 shall be atomized into each exhaust opening. Openings shall then be sealed with tape conforming to type II of MIL-T-22085. The engine crankcase breathers shall then be sealed with plastic plugs conforming to MIL-C-5501 or with tape conforming to type II of MIL-T-22085.

3.4.5.5.5 Preservation through dipstick shroud opening and oil filler tube. After preservation in accordance with 3.4.5.5.1 through 3.4.5.5.4, six ounces of preservative oil MIL-P-46002, grade 1, shall be atomize-sprayed into the crankcase through the oil filler cap opening. An extension of sufficient length to permit the spray nozzle to be within the crankcase shall be used. The spray nozzle shall not be submerged in the crankcase oil. In case of inability to use the oil tube filler system to atomize spray due to

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change in location of the filler tube, the dipstick shroud opening shall be used. After spraying has been accomplished, the dispstick shall be reinstalled, the oil filler cap closed and all openings to the engine interior, including dispstick shroud opening and oil filler cap, shall be sealed with tape, type II, MIL-T-22085.

3.4.5.5.6 Engine preservation tag. After processing in accordance with 3.4.5.5.3 through 3.4.5.5.5 a red tag shall be imprinted with the following warning: "ENGINE PRESERVED WITH VCI - DO NOT CRANK. BEFORE CRANKING ENGINE, REMOVE TAPE OR PLUGS FROM TURBOCHARGER INLETS, AIR CLEANER HOSES, EXHAUST TUBE OPENINGS, CRANKCASE BREATHERS, OIL FILLER CAP, DIPSTICK SHROUD OPENING, AND FROM OTHER OPENINGS TO THE ENGINE INTERIOR. RECONNECT AIR CLEANER HOSES TO TURBOCHARGER INLETS". Place the tag in a conspicuous location in the driver's compartment. Annotate DD Form 1397 to show that the engine has been preserved with VCI and preservative oil (see 6.2).

3.4.5.6 Air cleaners. After completion of acceptance tests and prior to chassis shipment, the air cleaner supply shall be adjusted to assure that air will be taken from the engine compartment and not the crew compartment. Exterior air cleaner discharge elbows shall be sealed with MIL-T-22085, type II.

3.4.5.7 Personnel heater and fuel pump. After processing the engine as specified in 3.1.7.5, uncouple the quick disconnect from the personnel heater fuel pump line and drain the fuel line. Seal the ends of the disconnected fuel line with plastic caps/plugs conforming to MIL-C-5501, or with tape conforming to type II of MIL-T-22085. The external heater exhaust opening shall be sealed with tape conforming to type II, MIL-T-22085. Warning tags with the following information shall be secured to the heater unit, to the heater fuel pump and to the heater operating switch (on driver's control panel): "HEATER FUEL LINE DISCONNECTED AND SEALED -REMOVE SEALS FROM FUEL LINE AND EXHAUST TUBE, OPERATE HEATER FUEL PUMP TO DRAIN MINIMUM OF ONE QUART FUEL - RECONNECT FUEL LINE TO HEATER PRIOR TO STARTING." (see 6.2).

3.4.5.8 Fuel tanks. Each fuel tank shall be drained to the maximum extent possible. The fuel tank cap and filler screen shall be removed and coated with lubricating oil conforming to type I, grade 30, of MIL-L-21260. One quart of lubricating oil conforming to type I, grade 10 of MIL-L-21260 shall be added to each five gallons, or portion thereof, of residual fuel. The tank cap and filler screen shall be reinstalled.

3.4.5.9 Ventilation. All access plates and gaskets on the underside of the chassis shall be removed, and the driver's and engine compartment drain valves secured in the open position. Unpainted, metal surfaces exposed by disassembly shall be coated with preservative conforming to Grade 1 of MIL-C-16173. Access plates shall be packaged in accordance with 3.1.8.4. Threaded portions exposed by removal of these items shall be coated with preservative conforming to grade 4 of MIL-C-16173. Screens conforming to figures 2 through 5 shall be constructed and installed in access cover openings as specified. The following shall be stenciled on the exterior of the vehicle: "REMOVE SCREENS, INSTALL ACCESS PLATES, COVERS, AND GASKETS, AND CLOSE DRAINS BEFORE OPERATING VEHICLE." Stenciling shall be applied using white or yellow paint conforming to MIL-P-13983. Characters shall be at least 3/4 inch in height (see 6.2).

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3.4.5.10 Fire extinguishers. Fire extinguisher cylinders shall be filled to at least 90 percent of rated full charge. All seals shall be intact. DA Form 253 shall be completed and securely attached to each cylinder (see 6.2 and 6.3).

3.4.5.11 Crew compartment escape hatch. Rubber seals around the crew compartment escape hatch shall be coated with powdered talc conforming to type IV, class C of ZZ-T-416. Hatch shall be secured from the inside.

3.4.5.12 Miscellaneous preservation. Except as otherwise specified herein, all exposed, unpainted, metal surfaces on the exterior of the chassis, except the track, shall be coated with compound conforming to grade 1 of MIL-C-16173. All exposed, unpainted, unplated ferrous metal surfaces on the interior of the chassis shall be coated with compound conforming to grade 4 of MIL-C-16173.

3.4.6 Packaging.

3.4.6.1 Dry charged batteries and cables. Dry charged batteries shall be installed in the vehicle battery carrier. Filler cap openings shall be sealed by placing a two inch wide by 3-mil thick strip of film conforming to type II of MIL-F-22191 over all filler cap openings with caps removed. The film shall be of sufficient length to allow the film to be depressed into the filler cap opening to the same depth as the filler cap. Filler caps shall be screwed into the filler openings to form a complete seal without damaging the plastic film. Battery cables shall be secured to the battery carrier with 3/4-inch wide tape conforming to AA-1685.

3.4.6.2 Electrolyte. Electrolyte shall be packaged, packed, and marked as specified for type IV, class 1 or 2 unit (as applicable) in accordance with O-S-801, except that the exterior containers shall conform to PPP-B-601 or PPP-B-621. The packed electrolyte shall be stowed with the basic issue items (BII) and secured independently to permit separate removal.

3.4.6.3 Fire extinguishers. Exterior fire extinguisher handles and protective shields shall be completely sealed with tape conforming to MIL-T-22085, type II. A red warning tag containing the following information shall be located in a conspicuous location within the driver's compartment. "EXTERIOR FIRE EXTINGUISHER HANDLES SEALED WITH TAPE - REMOVE TAPE BEFORE STARTING ENGINE OR PLACING VEHICLE IN SERVICE."

3.4.6.4 Access plates and gaskets. Preserved access plates and gaskets (see 3.4.5.9) shall be packaged in a box conforming to type CF, class WR, of PPP-B-636. The box shall be closed with tape conforming to type III, class 1 of PPP-T-60, identified as to contents, and securely stowed within the vehicle.

3.4.6.5 Tow hooks. Tow hooks and related hardware shall be removed for shipment and packaged in a type CF, class WR, box conforming to PPP-B-636. The box shall be closed with tape conforming to type III, class 1, of PPP-T-60, identified as to contents, and securely stowed within the vehicle.

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3.4.6.6 Basic issue items (BII). The basic issue items shall be packaged, packed, and stowed in accordance with MIL-P-12841 or other documents designated by the responsible agency. Unless otherwise specified, for shipping purposes, on-vehicle equipment shall be placed on the floor of railroad cars and blocked, braced and anchored to prevent movement in transit. Corner protectors shall be used under the strapping.

3.4.7 Chassis covers.

3.4.7.1 Closure kit. Unless otherwise specified (see 6.1), the chassis shall be provided with a protective closure. Closure for M60A1 chassis shall be fabricated, assembled, and installed in accordance with Drawing, K10893902. Closure for M48A5 chassis shall be fabricated, assembled, and installed in accordance with requirements specified in M60PD-T-62226(AT). All sharp corners of framework shall be cushioned with a 3/4 inch minimum t-thickness of cushioning material conforming to type I class 2 of PPP-C-1752 or type I of PPP-C-1797 secured in place with tape conforming to type II of MIL-T-22085. To prepare chassis for loading, front and rear stiffener rods shall be removed from cover. The cover shall be rolled away from the front and rear bows to expose chassis lifting eyes. The unused opening in the front of the closure cover shall be sealed according to figure 6.

3.4.7.1.1 Closure marking. The information "TO PREPARE VEHICLE FOR LOADING: OPEN ZIPPERS, REMOVE STIFFENER RODS NEAR COVER ENDS, UNFASTEN COVER FROM FRAME ENDS, AND ROLL COVER AWAY FROM FRONT AND REAR FRAME BOWS TO EXPOSE VEHICLE LIFTING EYES. AFTER LOADING, RESTORE AND SECURE COVER TO ORIGINAL CONDITION" shall be stenciled on the exterior front and rear of the closure in characters as minimum of 3/4 inch high using white enamel conforming to TT-E-529.

CLOSURE PACKAGING AND SHIPPING INSTRUCTIONS:

1. DISASSEMBLE FRAMEWORK AND SECURELY BUNDLE LIKE ITEMS.
2. PACKAGE SMALL HARDWARE IN CLOTH BAGS.
3. GROUP LARGEST, HEAVIEST ITEMS ON BOTTOM OF WOOD SHIPPING BOX.
4. PLACE SMALLER PACKAGED ITEMS IN VOIDS BETWEEN LARGER ITEMS.
5. FOLD CLOSURE COVER - PLACE ON TOP OF OTHER PACKED ITEMS.
6. IMMOBILIZE PACKED ITEMS AND SECURE BOX COVER.
7. SHIP TO (address to be furnished by contracting offficer.)

All stenciled characters shall be at least 3/4 inches high using white enamel conforming to TT-E-529.

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3.4.7.1.2 Closure disposition marking. The following information shall be stenciled on the outside, front and rear of the cover: "REUSABLE CLOSURE (COVER AND FRAMEWORK) - DO NOT DESTROY - WHEN REMOVED AND NO LONGER REQUIRED FOR VEHICLE PROTECTION, DISASSEMBLE, PACKAGE, AND SHIP PER INSTRUCTIONS ON INSIDE OF COVER." The following information shall be stenciled on the inside, front and rear, of the cover:

CLOSURE PACKAGING AND SHIPPING INSTRUCTIONS:

1. DISASSEMBLE FRAMEWORK AND SECURELY BUNDLE LIKE ITEMS.
2. PACKAGE SMALL HARDWARE IN CLOTH BAGS.
3. GROUP LARGEST, HEAVIEST ITEMS ON BOTTOM OF WOOD SHIPPING BOX.
4. PLACE SMALLER PACKAGED ITEMS IN VOIDS BETWEEN LARGER ITEMS.
5. FOLD CLOSURE COVER - PLACE ON TOP OF OTHER PACKED ITEMS.
6. IMMOBILIZE PACKED ITEMS AND SECURE BOX COVER.
7. SHIP TO (address to be furnished by contracting officer.)

All stenciled characters shall be at least 3/4 inches high using white enamel conforming to TT-E-529.

3.4.7.2 Manifold opening. The manifold opening shall be provided with covers constructed from plywood in accordance with figure 8. Plywood covers shall be placed over the manifold opening and secured with a bolt in accordance with figure 8. The joint between bolt head and cover, and hull and the cover, shall be sealed with tape conforming to type II of MIL-T-22085.

3.4.7.3 Crew compartment. Crew compartment opening shall be provided with a cover (see figure 7), constructed from plywood, a minimum of 1/2 inch thick, conforming to type I, group A of NN-P-530. Prior to installation of the cover, one of the access plates between the crew compartment and engine compartment shall be removed and secured within the crew compartment in a manner to prevent movement in transit. Eight wood spacers shall be placed over the bolt holes between the cover and "X" member of the frame. A plywood cover shall be placed over the crew compartment opening and secured by passing each bolt through the plywood and spacer then fastening to the existing bolt holes in the "X" members of the frame. A flat washer shall be used under the head of each bolt. The joint between the plywood cover and the hull shall be sealed with tape conforming to type II of MIL-T-22085. A 4-inch square screen constructed of wire cloth, 0.047 inch diameter, 4 by 4 mesh, conforming to type I, class I, or RR-W-360 shall be installed over the 3-inch diameter ventilation hole. The screen shall be secured to the cover with steel staples.

3.5 Level B. Chassis shall be processed in the same manner as specified for Level A, with the following exceptions.

3.5.1 Transmission and final drives. Transmission and final drives shall contain normal seasonal operational lubricant as specified in lubrication order, filled to operating level. DD Form 1397 shall be annotated to indicate grade of lubricant used (see 3.4.5.3).

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3.5.2 Engine crankcase. Engine crankcase shall contain normal seasonal operational lubricant as specified on lubrication order, filled to operational level. DD Form 1397 shall be annotated to indicate Grade of lubricant used (see 3.4.5.4).

3.5.3 Engine preservation. The engine shall not require preservation for Level B shipment and storage (see 3.4.5.5).

3.5.4 Air cleaners. The air cleaners shall neither be disassembled nor sealed for Level B shipment storage (see 3.4.5.6).

3.5.5 Fuel tanks. Unless otherwise specified (see 6.1), chassis shall be shipped without draining residual fuel from the fuel tanks (see 3.4.5.8).

3.5.6 Personnel heater and fuel pump. Unless otherwise specified, personnel heaters and fuel pumps shall be in a ready-to-use condition (see 3.4.5.7).

3.5.7 Closure kit. Vehicle closures shall not be provided on chassis processed for Level B shipment and storage (see 3.4.7.1).

3.5.8 Ventilation. All access plates and gaskets on the underside of the chassis shall be removed and exposed surfaces shall be preserved (see 3.4.5.9). Removed items shall be processed in accordance with 3.4.5.9. Both drain valves shall be secured in the open position. The opening in the crew compartment cover shall not be screened (see figure 10) but 3-inch hole and the edges between the wood cover and the hull shall be sealed with tape conforming to type II of MIL-T-22085. Surfaces to which tape will be applied shall be thoroughly cleaned and dried to assure effective tape adhesion. Access plates between the engine and crew compartments shall not be removed.

3.5.9 Cargo straps and fender boxes. Fabric retaining straps in the fender boxes shall be removed, identified with their Army part numbers, and be placed in a plastic bag conforming to type II, class B, style 2 of MIL-B-117. The bag shall be closed and stowed inside the chassis. Fender box covers shall be closed, and handles shall be locked and secured with a suitable gage wire conforming to QQ-W-461.

3.6 Loading.

3.6.1 Loading on flat cars. Loading of chassis on open top railcars shall be in accordance with the applicable requirements of Section 1, Association of American Railroads Manual, "Loading of Commodiites on Open Top Cars", and Figure Number 80 of Section 6 of the AAR rules, "Loading of Department of Defense Material on Open Top Cars." The number of units to be loaded on each railcar, the type of railcar, and the applicable transportation data shall be as authorized by the responsible government transportation office. (See 3.1.9.1 for preparing closure covers for loading activities.)

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3.6.1.1 Lifting and tiedown devices. When specified (see 6.2), the vehicle shall be provided with slinging and tiedown provisions in accordance with MIL-STD-209 and MIL-A-8421, or the location of hard points for attachment of lifting and tiedown devices shall be specified (see 3.4.1).

3.6.2 Reprocessing engine after loading - Level A. If the engine is operated in connection with moving the chassis to loading area, or during chassis loading or unloading, the engine shall be reprocessed as specified in 3.1.7.5. The closure cover shall be rolled clear of the engine intake and exhaust to provide air circulation and to prevent damage to the cover. After reprocessing, the closure shall be replaced in its original position.

3.6.2.1 Reprocessing engine after loading - Level B. If the engine is operated in connection with movement of the chassis during loading or unloading, additional processing of the engine will not be required.

3.7 Marking. In addition to any special marking required in the contract or order, the chassis shall be marked in accordance with 3.4.1 and MIL-STD-129.

3.7.1 Lifting points. The legend "LIFT HERE" with an arrow pointing to the lifting eye shall be stenciled adjacent to each lifting eye using black enamel conforming to TT-E-527, No. 37038. Stenciling shall be at least 3/4 inch high.

3.7.2 Shipping label adhesion. To ensure effective adhesion when applied during cold weather, Military Shipment Labels, DD Form 1387, shall be cemented to vehicles with adhesive conforming to type I of MMM-A-1617. After mounting, labels shall be covered with a protective coating in accordance with applicable provisions of MIL-STD-129.

3.8 Drive on - drive off capability. When chassis are to be operated for loading or unloading (see 6.1), the following provisions shall apply.

3.8.1 Fuel tanks. Additional fuel shall be added, as required, to accomplish movement of chassis.

3.8.2 Batteries and electrolyte. Batteries shall be filled with electrolyte and be fully charged, and cables shall be properly connected (see 3.4.6.1 and 3.4.6.2). After chassis self-movement for loading or for placement in storage, the main power lead to the master control box in the driver's compartment shall be disconnected and secured to prevent movement. A tag bearing the following message, "VEHICLE PRESRVED FOR DRIVE-AWAY CONDITION. BEFORE CRANKING, CONNECT HARNESS (CIRCUIT 81) TO MASTER RELAY BOX ON HULL FLOOR UNDER TURRET BASKET. ENGINE AND FUEL TANKS NOT PRESERVED" shall be placed in a conspicuous location in the crew compartment.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform or witness any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection records. Contractor shall maintain records of all inspections performed and such records shall be readily availabe for reveiw by the Government representative.

4.2 Classification of inspection.

a. First article inspection (see 4.3).
b. Quality conformance inspections (see 4.4).

4.3 First article inspection. Unless otherwise specified (see 6.2), the Government shall select one of the first ten processed tank chassis produced under the production contract for first article inspection. First article samples shall be inspected for conformance to the requirements listed in table I. Approval of the first article sample by the Government shall not relieve the contractor of his obligation to supply processed tank chassis that are fully representative of those inspected as a first article sample. Any changes or deviation of the production units from the first article sample shall be subject to the approval of the contracting officer.

4.3.1 First article inspection failure. Test item deficiences during, or as a result of, the first article test, shall be cause for rejection of the items until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiency. Any deficiency found during, or as a result of the first article test, shall be prime facie evidence that all items already produced prior to completion of the first article test are similarly deficient unless evidence satisfactory to the contracting officer is furnished by the contractor that they are not similarly deficient. Such deficiencies on all items shall be corrected by the contractor at no cost to the Government. The Government shall not final accept products until first article testing is completed to the satisfaction to the Government."

4.4 Quality conformance inspection.

4.4.1 Quality conformance inspection.. One vehicle each day shall be inspected in accordance with 4.5.2.1. Each vehicle shall be inspected in accordance with 4.5.2.2 through 4.5.2.5 and for conformance to the requirements of Table I.

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4.4.2 Failure. Failure of any processed tank chassis to pass any of the specified tests shall be cause for the Government to refuse acceptance of the production quantity represented, until action taken by the contractor to correct defects and prevent recurrence has been approved by the Government.

4.5 Methods of inspection.

4.5.1 Materials. To insure conformance to 3.2, except for materials which have been inspected by the Government at source, all materials to be used in processing of vehicles shall be inspected in accordance with the material specification; or certified inspection and test reports shall be provided which show that furnished materials conform to the detailed specifications. When materials are listed on a Qualified Products List, they shall be obtained from one of the approved sources indicated.

4.5.2 Processing. To ensure conformance 3.3 through 3.4.3 and 3.5, inprocess inspection shall include all items specified in table I and 4.5.2.1 through 4.5.2.3.

4.5.2.1 Cleaning. To determine conformance to 3.4.4.2, interior of vehicles shall be examined for cleanliness. One vehicle each day shall be tested for cleanliness in accordance with the applicable provisions of MIL-P-116. To determine conformance to 3.4.4.3, exterior of vehicle shall be examined for cleanliness. Surfaces to which tape is to be applied shall be examined for cleanliness before application (see 3.5.9).

4.5.2.2 Fuel tanks. To determine conformance to 3.4.5.8, the fuel tank interior shall be visually inspected to verify that specified processing has been accomplished.

4.5.2.3 Engine. To determine conformance to 3.4.5.5, interior of the engine from the first processed chassis shall be examined for surface coverage. One cylinder head shall be removed to permit visual examination of surfaces within the combustion chamber. Surfaces within the combustion chamber, including piston crown, cylinder wall and chamber head, shall have a "wet" coating or preservative oil such as is obtained when an item is dipped or flushed with the oil. The processing method used to prepare the approved preserved engine shall be applied to subsequent production chassis (see 3.1).

5. Preparation for delivery. This section is not applicable to this specification.

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TABLE I. Classification of requirements.

(See Indicated paragraphs for Level A and B requirements)

Component or Processing Activity	Cleaning	Preservation		Packaging/ Stowage
	Levels A & B	Level A	Level B	Levels A & B
Disassembly		3.4.1		3.4.1
Matchmarking				3.4.2
Record forms				3.4.3
Cleaning and drying	3.4.4			
Interior of vehicle	3.4.4.1			
Battery supports & retainers	3.4.4.2			
Exterior of vehicle	3.4.4.3			
Preservation		3.4.5	3.4.5	
Relubrication		3.4.5.1	3.4.5.1	
Battery supports & retainers		3.4.5.2	3.4.5.2	
Transmission & final drives		3.4.5.3	3.5.1	
Engine crankcase		3.4.5.4	3.5.2	
Engine preservation		3.4.5.5	3.5.3	
Preservation through fuel system and combustion chamber		3.4.5.5.1	3.5.3	
Engine purging		3.4.5.5.2	3.5.3	
Preservation of turbochargers		3.4.5.5.3	3.5.3	
Preservation through exhaust system		3.4.5.5.4	3.5.3	
Preservation through dipstick shroud opening & oil filler tube		3.4.5.5.5	3.5.3	
Engine preservation warning tag		3.4.5.5.6	3.5.3	
Air cleaners		3.4.5.5.7	3.5.4	
Personnel heater & fuel pump		3.4.5.5.8	3.5.6	
Fuel tanks		3.4.5.5.9	3.5.5	
Ventilation		3.4.5.5.10	3.5.8	

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TABLE I. Classification of requirements - Continued.

(See Indicated paragraphs for Level A and B requirements)

Component or Processing Activity	Cleaning	Preservation		Packaging/ Stowage
	Levels A & B	Level A	Level B	Levels A & B
Fire extinguishers		3.4.5.5.10		
Escape hatch		3.4.5.5.11		
Miscellaneous		3.4.5.5.12		
Packaging				
Dry charged and batteries & cables				3.4.6.1
Electrolyte				3.4.6.2
Fire extinguishers				3.4.6.3
Access plates, gaskets & drain plugs				3.4.6.4
Tow hooks				3.4.6.5
Basic issue items (BII)				3.4.6.6
Closure kit		3.4.7.1	3.5.7	
Manifold opening				3.4.7.2
Crew compartment cover				3.4.7.3
Cargo straps & fender boxes (Level B only)				3.5.9
Loading on flat cars				3.6.1
Lifting and Tiedown Devices				3.6.1.1
Reprocessing engine after loading-				
Level A				3.6.2
-Level B				3.6.2.1
Marking				3.7
Lifting points				3.7.1
Shipping label adhesion				3.7.2
Drive on-drive off				3.8

6. NOTES

6.1 Intended use. This specification covers processing of M60A1 and M48A5 vehicle for storage outside of buildings, for immediate shipment and domestic or overseas shipment including carloading (see 1.1).

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Applicable level type of processing required (see 1.2).
c. If first article samples are required (see 3.1).
d. If vehicle closure is required (see 3.4.7.1).
e. If additional fuel shall be supplied (see 3.8.1).
f. If chassis drive on/drive off capability is required (see 3.8).
g. Whether slinging or tiedown devices are to be provided (see 3.6.1.1).
h. If special marking is required (see 3.4.5.4, 3.4.5.5.6, 3.4.5.7, 3.4.5.9, 3.4.5.10, 3.4.6.3 and 3.4.1.1.1).
i. If responsibility for inspection shall be other than as specified (see 4.1).

6.3 Safety precautions. Caution should be exercised in handling carbon dioxide (CO_2) fire extinguisher cylinders. Cylinders should not be dropped, be permitted to strike each other, or be handled roughly. Extreme care should be exercised during the reinstallation operaton to avoid tripping fire extinguisher control system (see 3.4.5.10).

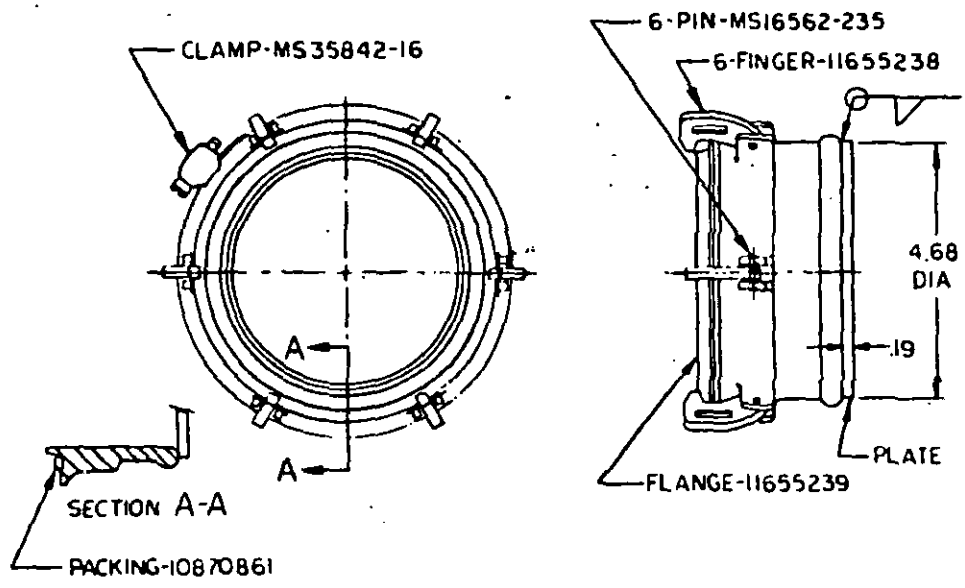
6.4 Forms. A copy of the "Equipment Log Book" and all required forms (see 3.4.3) will be furnished to the contractor by the Government at least 30 days before shipment of the vehicles required by the contract delivery schedule.

Custodian:
Army - AT

Preparing activity:
Army - AT

(Project No. 2350-A313)

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REQUIREMENTS

MATERIALS:

- Flange, Fingers, Pins, Clamp, and Packing: Requirements as specified on Army and Military Standard drawings indicated on SH 1.

- Plate: Aluminum plate 6061, 5083 or 5456, Spec. QQ-A-250.

WELDING:

- Weld Plate to Flange per MIL-W-45205, Class A. Use aluminum filler conforming to MIL-E-16053 or QQ-R-566. Air-tight joints required.

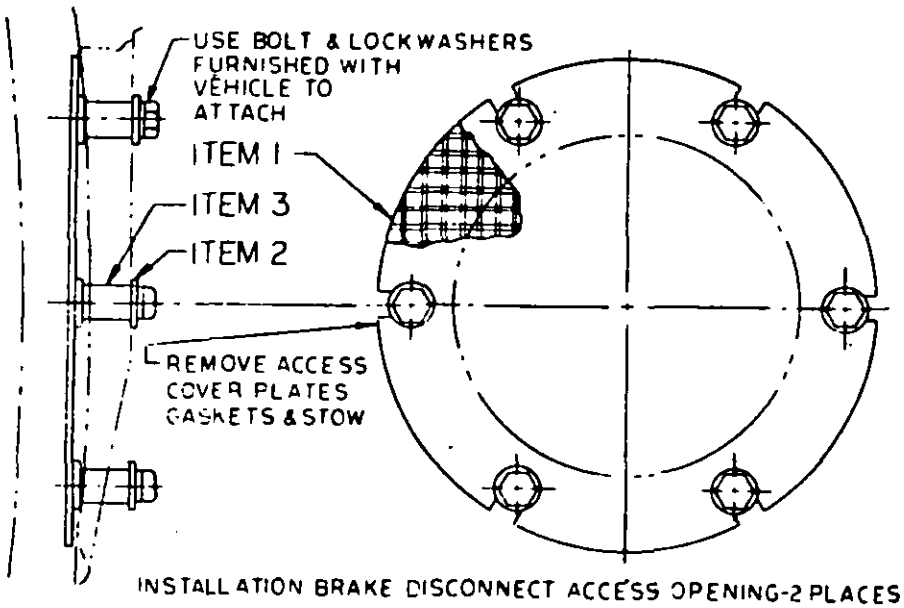
BONDING:

- Bond Packing to Flange with adhesive MMM-A-1617, Type II. Cured bond to withstand 5-lb pull per inch of width perpendicular to bonded surface.

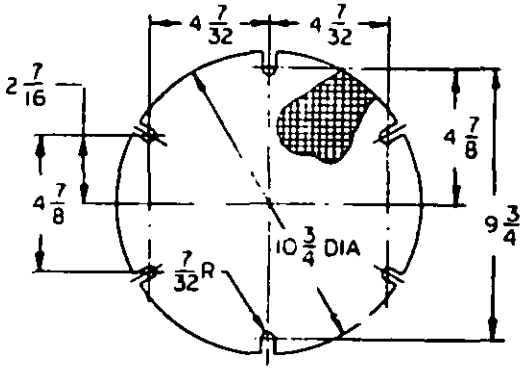
TURBOCHARGER AIR INLET RESTRICTOR COVER

Figure 1

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INSTALLATION BRAKE DISCONNECT ACCESS OPENING-2 PLACES



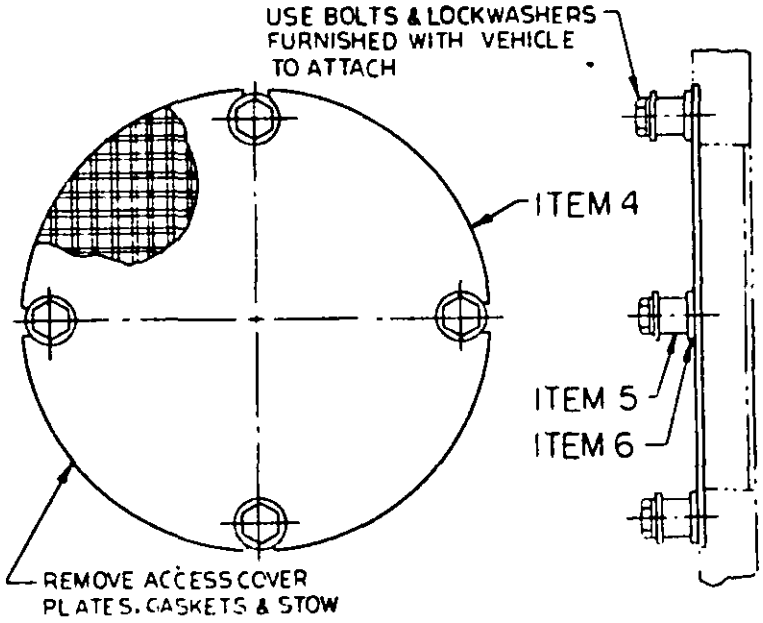
ITEM NO. 1

INSTALLATION SCREENS IN BRAKE ACCESS OPENING

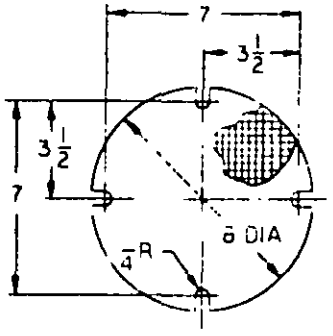
ITEM NO.	FIG. NO.	NO. REQ'D	NAME	MATL	STOCK SIZE
1	B2	2	SCREEN	STEEL	GALVANIZED WIRE .047 DIA, 4X4 MESH
2	—	24	WASHER	STEEL	3/8 I.D. PLAIN
3	—	12	SPACER	STEEL TUBING	9/16 O.D. X .083 WALL X 7/8 LG.

Figure 2

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INSTALLATION-FUEL TANK DRAIN ACCESS OPENING
TWO PLACES



ITEM NO. 4

INSTALLATION SCREENS IN FUEL TANK ACCESS OPENING

ITEM NO.	FIG. NO.	NO. REQD	NAME	MATL	STOCK SIZE
4	B3	2	SCREEN	STEEL	GALVANIZED WIRE 047 DIA, 4X4 MESH
5	—	8	SPACER	STEEL TUBING	11/16 O.D. X .065 WALL X 9/16 LG.
6	—	16	WASHER	STEEL	1/2 I.D. PLAIN

Figure 3

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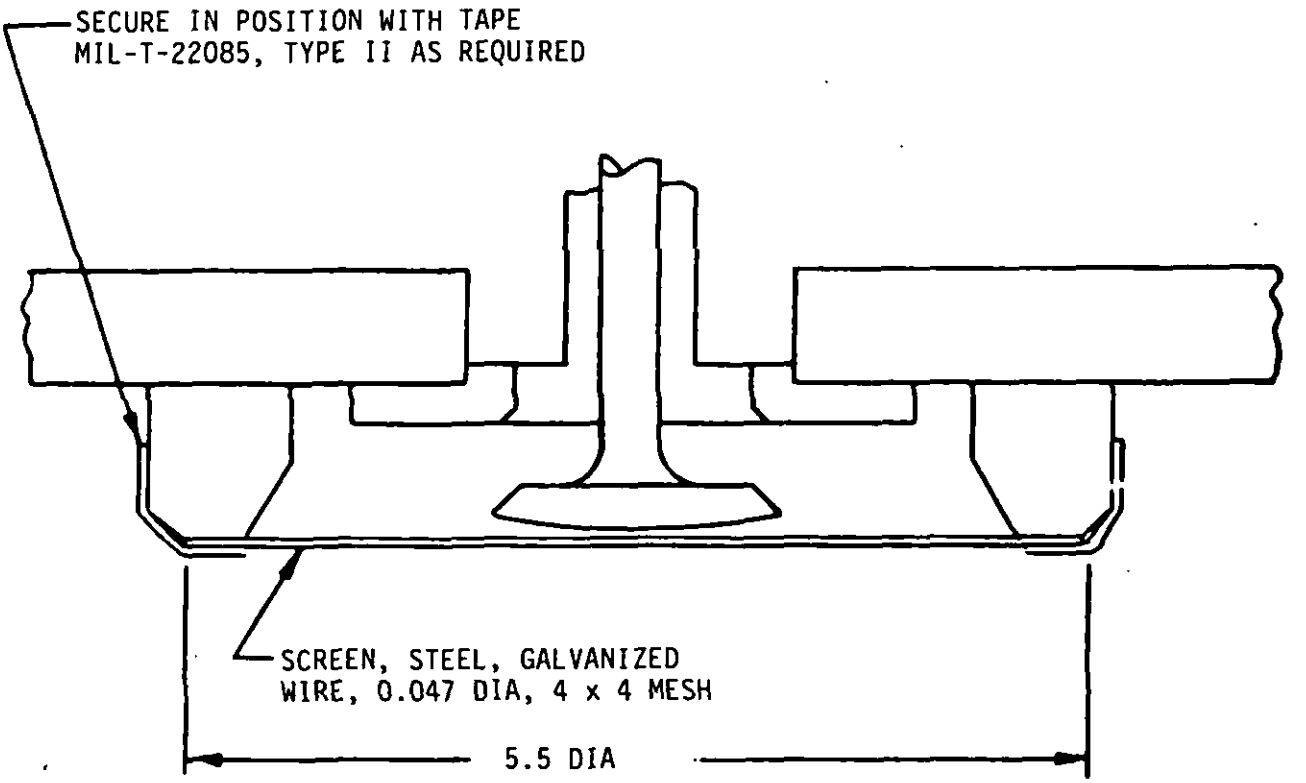


FIGURE 4. Installation, driver compartment drain valve screen.

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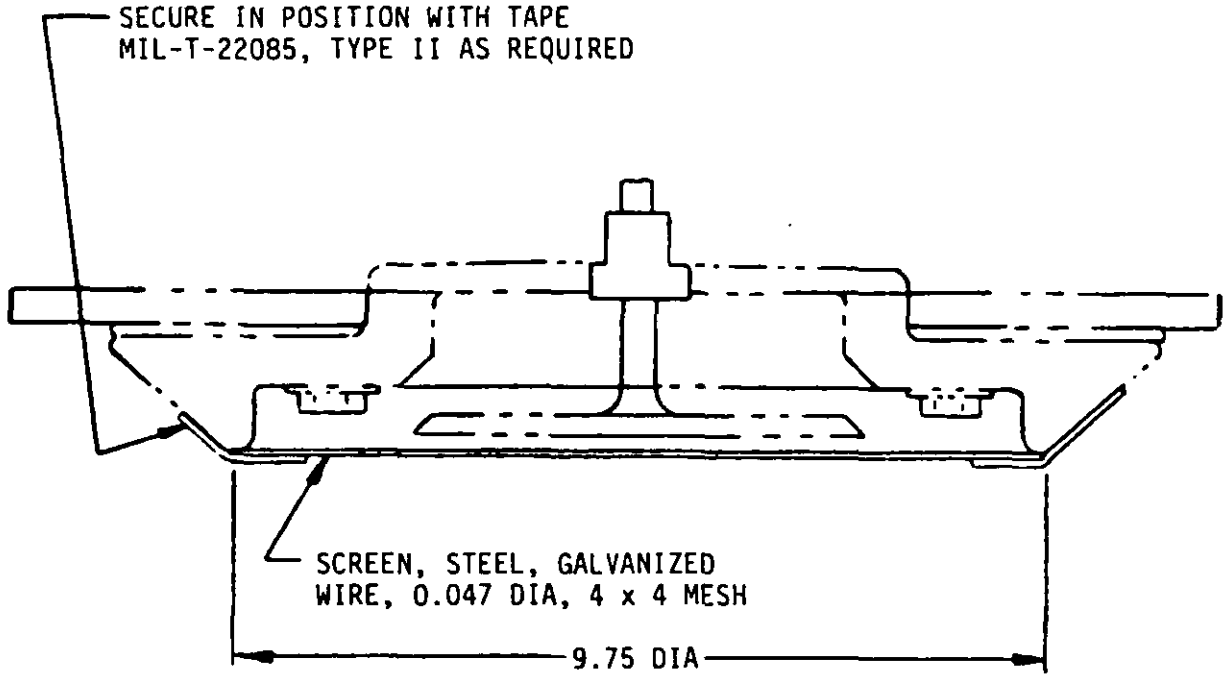
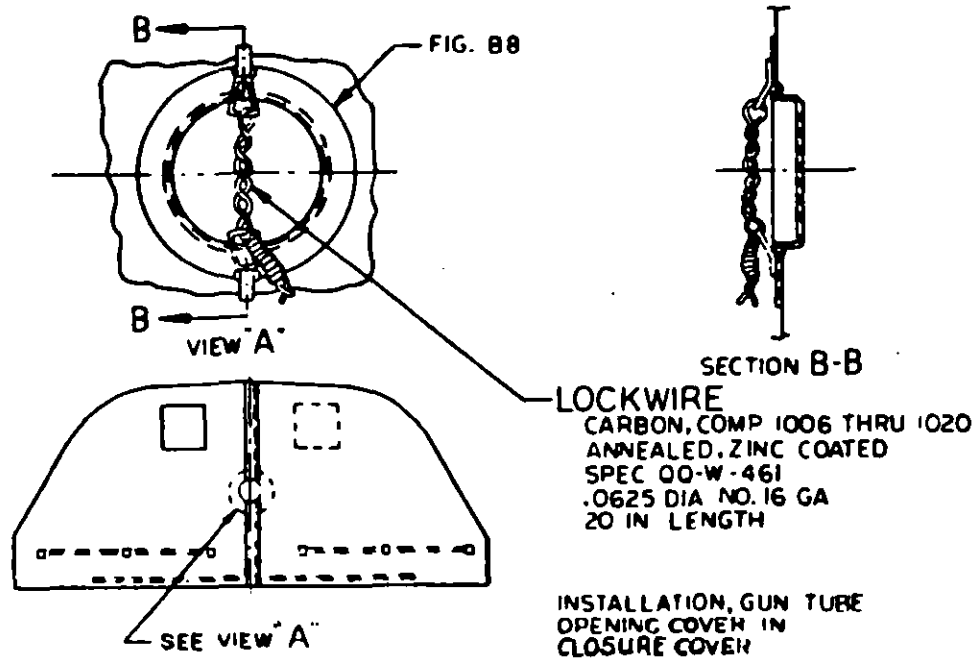
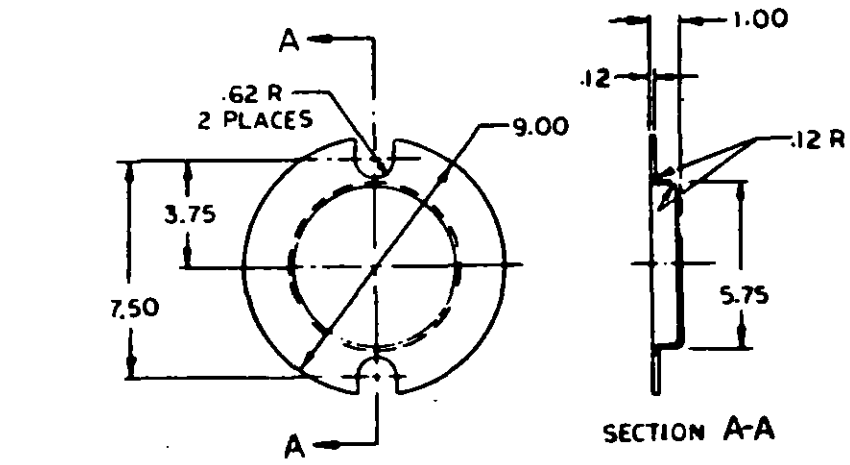


FIGURE 5. Installation engine compartment drain valve screen.

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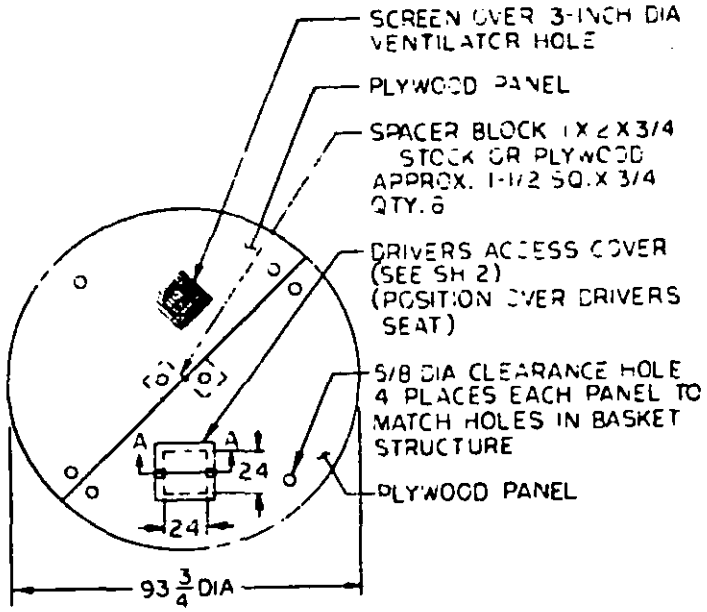




MATERIAL:
COVER, NYLON
TYPE I, SPEC MIL-L-20693
COLOR O.D.
TOLERANCES ± .06

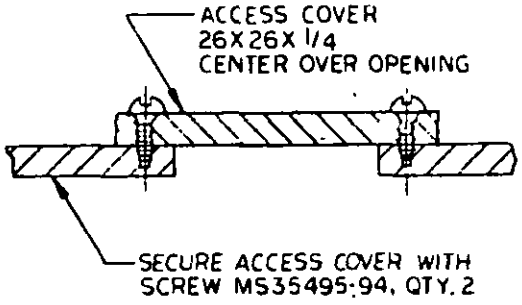
Figure 6

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NOTE:
SECURE PLYWOOD PANELS TO BASKET STRUCTURE WITH
SCREW MS90726-114
WASHER MS27183-18
QTY. 8 EACH

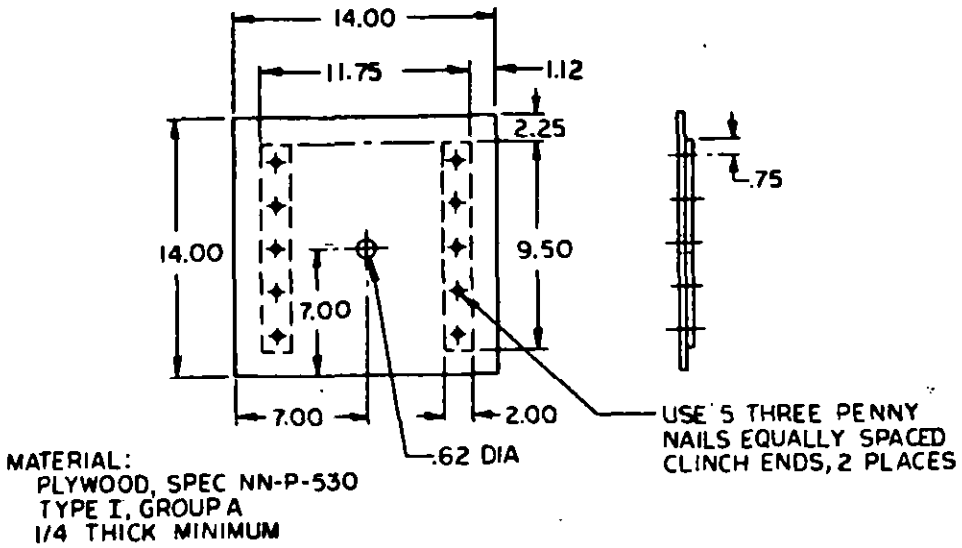
MATERIAL:
PLYWOOD,
SPEC NN-P-530
TYPE I, GROUP A
1/2 THICK MIN.



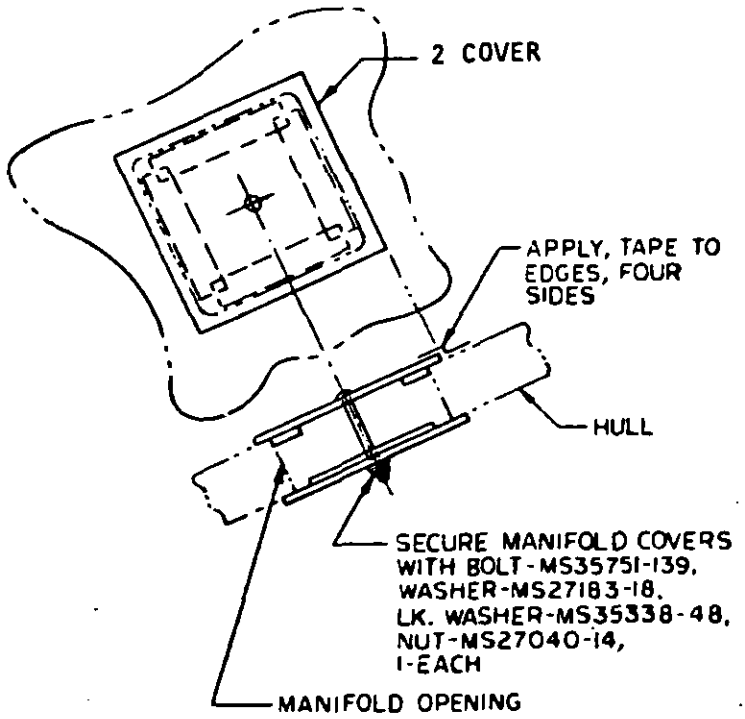
CREW COMPARTMENT COVER

Figure 7

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MANIFOLD OPENING COVER (2 REQD)



MANIFOLD COVER INSTALLATION

Figure 8

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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